Akash Kumar

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Education

VIT Bhopal University **2023 – 2027**
Bachelor of Technology, Computer Science & Engineering | CGPA: **7.76/10.0** Bhopal, India

Technical Skills

AI/ML: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB)
Languages: C++, Python, JavaScript, SQL
Web & Frameworks: React.js, Node.js, REST APIs
Databases: PostgreSQL, Supabase (DB & Auth), MySQL
Cloud & DevOps: AWS, Docker, Vercel, Render
Tools & Practices: Git, GitHub

Projects

Real-Time Collaborative Whiteboard | **GitHub** Mar 2026 – Apr 2026

- Teams lacked a shared canvas – orchestrated an **event-driven** sync layer using **WebSockets (Socket.IO)** to broadcast board state – achieved low-latency collaboration for **5+ concurrent users**.
- Dropped connections risked lost strokes – hardened **reconnection logic** and **failure-case handling**, and documented architecture trade-offs – kept sessions consistent for **5+ users** and the codebase maintainable.
- Interface iterations moved slowly – modularized the UI into reusable **React.js** components styled with **Tailwind CSS** through **Agile** sprints – accelerated UX refinements with every release.

AI-EchOo Grocery Platform | **Live Demo** | **GitHub** May 2026 – Jun 2026

- Shoppers faced a static catalog – built a **generative-AI** shopping assistant across **6 purchase flows** – turned browsing into guided discovery and comparison.
- Assistant answers needed real-time context – designed a relational **4-table** schema and integrated **Groq**-hosted **LLMs** with **RAG** – anchored responses in **live inventory**.
- Shoppers needed hands-free ordering – implemented **Web Speech Recognition** for voice commands with instant transcription – enabled natural, conversational shopping via automatic query submission.

Interview IQ | **Live Demo** | **GitHub** Jul 2025 – Aug 2025

- Candidates lacked reliable prep – architected a document-grounded **RAG** pipeline with **LangChain** and a **vector database** for **context-aware retrieval** – surfaced verified, source-backed answers.
- Generic guidance limited personalization – engineered a **PostgreSQL**-backed data layer that logged history across **4 activity types** – tailored each candidate's study path with real signals.
- Scattered resources slowed study – shipped **3 formats** of **LLM**-generated study material, **VAPI**-powered **voice AI** interviews, and a **resume gap analyzer** – unified preparation into one workflow.

Achievements

- **InnovateNSUT Hackathon** (Link) – Competed against **250 teams** in a team hackathon held at NSUT Delhi – placed **5th** overall.
- **Codeforces** (Link) – Reached a rating of **1124** in rated online contests – recorded global ranks of **5097**, **5972**, and **6035**.
- **CodeChef** (Link) – Earned **2 Star** status at **1459** rating in timed coding rounds – ranked **483rd**, **740th**, and **1078th** worldwide.
- **Competitive Coding** – Practiced **data structures** and **algorithms**, solved **400+ problems**.

Certifications

- **Cybersecurity Analyst** (Link) – IBM, threat detection, incident response, security tools.
- **Advanced Software Engineering & Development Internship** (Link) – MP Online.
- **Data Analyst Certificate** (Link) – Google, SQL, data cleaning, visualization.